

PMATH 465 Notes

Smooth Manifolds

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1 Smooth Manifolds

1.1 Coordinate Charts

The concept of a manifold starts with the notion of coordinate charts.

Definition 1.1 (Coordinate chart). A *coordinate chart* on a set X is a subset $U \subseteq X$ together with a bijection

$$\varphi : U \longrightarrow \varphi(U) \subseteq \mathbb{R}^n,$$

where $\varphi(U)$ is open in $\mathbb{R}^n$. The map φ allows us to parametrize points $x \in U$ by coordinates

$$\varphi(x) = (x_1, \dots, x_n).$$

Example 1.2. For $X = \mathbb{R}^n$, take $U = X$ and $\varphi = \text{id}_{\mathbb{R}^n}$. More generally, if $U \subseteq \mathbb{R}^n$ is open, then $\varphi = \text{id}_U$ gives a chart on U .

Example 1.3 (Graph of a function). Let $U \subseteq \mathbb{R}^n$ be open and let $f : U \rightarrow \mathbb{R}^m$. Define

$$X = \{(x, f(x)) : x \in U\} \subseteq \mathbb{R}^{n+m}.$$

Then

$$\varphi : X \rightarrow U, \quad (x, f(x)) \mapsto x$$

is a coordinate chart on X .

Example 1.4 (The circle). Let

$$S^1 = \{(x_1, x_2) \in \mathbb{R}^2 : x_1^2 + x_2^2 = 1\}.$$

Using two open pieces $U_1 = S^1 \setminus \{(-1, 0)\}$ and $U_2 = S^1 \setminus \{(1, 0)\}$, one may use stereographic-coordinate-type charts. For example,

$$\varphi_1(x_1, x_2) = \frac{x_2}{1 + x_1}, \quad \varphi_2(x_1, x_2) = \frac{x_2}{1 - x_1}.$$

Each map identifies the corresponding open arc with an open subset of $\mathbb{R}$.

Example 1.5 (The set of straight lines in $\mathbb{R}^2$). Let X be the set of all

straight lines in $\mathbb{R}^2$. A nonvertical line has a unique equation

$$y = mx + b,$$

so the nonvertical lines are parametrized by $(m, b) \in \mathbb{R}^2$. Similarly, a non-horizontal line can be written in a second coordinate form. These two parameterizations provide two coordinate charts.

Definition 1.6 (Smooth atlas). A smooth n -dimensional atlas on a set X is a collection of coordinate charts

$$\{(U_\alpha, \varphi_\alpha)\}_{\alpha \in A}$$

such that:

1. $X = \bigcup_{\alpha \in A} U_\alpha$;
2. for every α, β , the set

$$\varphi_\alpha(U_\alpha \cap U_\beta)$$

is open in $\mathbb{R}^n$;

3. whenever $U_\alpha \cap U_\beta \neq \emptyset$, the transition map

$$\varphi_\beta \circ \varphi_\alpha^{-1} : \varphi_\alpha(U_\alpha \cap U_\beta) \longrightarrow \varphi_\beta(U_\alpha \cap U_\beta)$$

is smooth.

Remark 1.7. Because the indices run through all pairs α, β , the inverse transition map also occurs. Hence the transition maps are diffeomorphisms between their domains.

Example 1.8. For $X = \mathbb{R}^n$, the single chart $(\mathbb{R}^n, \text{id})$ is a smooth atlas. More generally, the collection of identity charts on open subsets gives the usual smooth structure on $\mathbb{R}^n$.

Example 1.9 (Another atlas on S^1). Let

$$U_1 = S^1 \setminus \{(0, -1)\}, \quad U_2 = S^1 \setminus \{(0, 1)\}.$$

Projection onto the x_1 -coordinate gives charts onto $(-1, 1)$ after choosing the appropriate upper/lower half. This produces a smooth atlas distinct from the stereographic one but compatible with it.

Definition 1.10 (Smooth manifold, preliminary form). An n -dimensional smooth manifold is a set that admits an n -dimensional smooth atlas.

Example 1.11 (Real projective space). Real projective space $\mathbb{R}P^n$ is the set of one-dimensional subspaces of $\mathbb{R}^{n+1}$, equivalently the set of lines through the origin.

For $v = (v_1, \dots, v_{n+1}) \neq 0$, write L_v for the line spanned by v . Set

$$U_i = \{L_v : v_i \neq 0\}.$$

Every $L_v \in U_i$ has a unique representative with i th coordinate equal to 1, and therefore a chart

$$\varphi_i : U_i \longrightarrow \mathbb{R}^n$$

obtained by deleting the i th coordinate of that representative. These charts form a smooth atlas on $\mathbb{R}P^n$.

1.2 Manifolds, Smooth Structures, and Topology

Definition 1.12 (Compatible atlases). Two smooth atlases are *compatible* if their union is again a smooth atlas.

Remark 1.13. Compatibility is an equivalence relation on smooth atlases.

Definition 1.14 (Smooth structure). A smooth or differentiable structure on a set X is an equivalence class of compatible atlases.

Definition 1.15 (Smooth manifold). A smooth manifold is a set together with a smooth structure.

Note. To show that a set X is an n -dimensional manifold, it is enough to construct one smooth n -dimensional atlas. The maximal atlas is the union of all charts compatible with the chosen smooth structure.

1.2.1 Topological preliminaries

Definition 1.16 (Topology). Let X be a set. A topology on X is a collection $\mathcal{T}$ of subsets of X satisfying:

1. $\emptyset, X \in \mathcal{T}$;
2. arbitrary unions of sets in $\mathcal{T}$ belong to $\mathcal{T}$;
3. finite intersections of sets in $\mathcal{T}$ belong to $\mathcal{T}$.

The elements of $\mathcal{T}$ are called open sets. A subset $F \subseteq X$ is closed if $X \setminus F$ is open.

Example 1.17. The collection $\{\emptyset, X\}$ is a topology. On $\mathbb{R}^n$, the standard topology is generated by open balls.

Definition 1.18 (Relative topology). If $(X, \mathcal{T})$ is a topological space and $A \subseteq X$, define

$$\mathcal{T}_A = \{A \cap U : U \in \mathcal{T}\}.$$

This is the relative or subspace topology on A .

Suppose M is a smooth n -manifold with smooth atlas $\{(U_\alpha, \varphi_\alpha)\}$. A subset $V \subseteq M$ is called open if $\varphi_\alpha(V \cap U_\alpha)$ is open in $\mathbb{R}^n$ for every α .

Proposition 1.19. The open sets defined by the atlas form a topology on M , called the natural topology.

Proof. For each chart, openness is tested after applying the coordinate map. Since coordinate images of unions are unions and coordinate images of finite intersections are finite intersections, the topology axioms follow from the corresponding axioms in $\mathbb{R}^n$. $\square$

Proposition 1.20. Each chart map

$$\varphi_\alpha : U_\alpha \rightarrow \varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$$

is a homeomorphism for the natural topology on M and the standard topology on $\mathbb{R}^n$.

Proof. By definition, φ_α carries open sets of U_α to open sets in $\varphi_\alpha(U_\alpha)$. Con-

versely, if $W \subseteq \varphi_\alpha(U_\alpha)$ is open, then for any overlapping chart (U_β, φ_β) ,

$$\varphi_\beta(\varphi_\alpha^{-1}(W) \cap U_\beta) = (\varphi_\beta \circ \varphi_\alpha^{-1})(W \cap \varphi_\alpha(U_\alpha \cap U_\beta)),$$

which is open because the transition map is a diffeomorphism. $\square$

1.2.2 Hausdorff and second countable

Definition 1.21 (Hausdorff). A topology $\mathcal{T}$ on X is Hausdorff if for any $p \neq q$ in X , there exist disjoint open sets U, V with $p \in U$ and $q \in V$.

Definition 1.22 (Basis). A basis $\mathcal{B}$ for a topology $\mathcal{T}$ on X is a collection of open sets such that:

1. for every $x \in X$, some $B \in \mathcal{B}$ contains x ;
2. if $x \in B_1 \cap B_2$ with $B_1, B_2 \in \mathcal{B}$, then there is $B_3 \in \mathcal{B}$ such that

$$x \in B_3 \subseteq B_1 \cap B_2.$$

Definition 1.23 (Second countable). A topology is second countable if it admits a countable basis.

The notes impose two conditions on an atlas that ensure the natural topology is Hausdorff and second countable:

- (i) countably many chart domains cover the manifold;
- (ii) if $p \neq q$, then either p, q lie in one chart domain, or there are chart domains containing them that are disjoint.

Lemma 1.24. Under the separation condition above, the natural topology is Hausdorff.

Proof. If p and q lie in disjoint chart domains, we are done. Otherwise they lie in one chart U_α . Since $\varphi_\alpha(U_\alpha)$ is Hausdorff, choose disjoint Euclidean open sets separating $\varphi_\alpha(p)$ and $\varphi_\alpha(q)$, then pull them back by φ_α . $\square$

Lemma 1.25. If countably many chart domains cover M , then the natural topology is second countable.

Proof. For each chart image $\varphi_\alpha(U_\alpha) \subseteq \mathbb{R}^n$, choose a countable basis. Pull all these basis elements back through the chart maps. A countable union of

countable collections is countable, and the resulting collection is a basis for the natural topology. $\square$

Definition 1.26 (Topological n -manifold). A topological n -manifold is a Hausdorff, second-countable topological space in which every point has an open neighborhood homeomorphic to an open subset of $\mathbb{R}^n$.

Definition 1.27 (Atlas on a topological manifold). An atlas for a topological n -manifold is a collection of coordinate charts $\{(U_\alpha, \varphi_\alpha)\}$ covering the space.

Definition 1.28 (Smooth atlas and smooth structure). An atlas is smooth if every transition map is smooth. A smooth structure is a maximal smooth atlas, and a topological manifold equipped with such a structure is a smooth manifold.

Example 1.29 (Line with doubled origin). Take two copies of $\mathbb{R}$ and identify $(x, a) \sim (x, b)$ whenever $x \neq 0$. The result carries a natural smooth atlas from the two copies of $\mathbb{R}$, but it is not Hausdorff: the two origins cannot be separated by disjoint open sets.

Example 1.30. A disjoint union of uncountably many copies of $\mathbb{R}$ is a manifold in the local sense but fails second countability.

1.3 Regular Level Sets

Theorem 1.31 (Regular level set theorem). Let $U \subseteq \mathbb{R}^{n+m}$ be open and let

$$F : U \rightarrow \mathbb{R}^m$$

be smooth. Fix $c \in \mathbb{R}^m$ and set

$$F^{-1}(c) = \{x \in U : F(x) = c\}.$$

If $DF_a : \mathbb{R}^{n+m} \rightarrow \mathbb{R}^m$ is surjective for every $a \in F^{-1}(c)$, then $F^{-1}(c)$ is a smooth n -dimensional manifold, Hausdorff and second countable.

Example 1.32 (Sphere). Let

$$S^n = \{x \in \mathbb{R}^{n+1} : x_1^2 + \cdots + x_{n+1}^2 = 1\}.$$

Take

$$F(x) = \|x\|^2.$$

Then

$$DF_x = [2x_1 \ \cdots \ 2x_{n+1}],$$

which is surjective whenever $x \neq 0$. In particular, every $x \in S^n$ is a regular point of F , so S^n is a smooth n -manifold.

Example 1.33 (Special linear group). Identify $M_{n \times n}(\mathbb{R})$ with $\mathbb{R}^{n^2}$ and consider

$$\det : M_{n \times n}(\mathbb{R}) \rightarrow \mathbb{R}.$$

For a fixed $c \in \mathbb{R}$, let

$$X_c = \{A : \det A = c\}.$$

The derivative satisfies

$$\frac{\partial \det A}{\partial a_{ij}} = (\operatorname{adj} A)_{ji}.$$

When $c \neq 0$, the matrix A is invertible, so $\operatorname{adj} A \neq 0$. Hence the derivative is nonzero and therefore surjective as a map to $\mathbb{R}$. Thus X_c is a smooth manifold of dimension $n^2 - 1$. In particular,

$$\operatorname{SL}(n, \mathbb{R}) = \{A : \det A = 1\}$$

is a smooth $(n^2 - 1)$ -manifold.

1.4 Smooth Maps Between Manifolds

Let M and N be smooth manifolds of dimensions m and n .

Definition 1.34 (Smooth map). A map $F : M \rightarrow N$ is smooth if for every $x \in M$, and for charts (U, φ) about x and (V, ψ) about $F(x)$ with $F(U) \subseteq V$, the coordinate representation

$$\psi \circ F \circ \varphi^{-1} : \varphi(U) \rightarrow \psi(V)$$

is smooth in the ordinary Euclidean sense.

Note. The condition $F^{-1}(V)$ open in M guarantees continuity with respect to the natural topologies. Smoothness may be checked

using any choice of compatible charts because a change of charts composes the coordinate expression with smooth transition maps.

Definition 1.35 (Diffeomorphism). A diffeomorphism $F : M \rightarrow N$ is a smooth bijection with smooth inverse.

Remark 1.36. Diffeomorphic manifolds have the same dimension.

Example 1.37. Every coordinate chart $\varphi : U \rightarrow \varphi(U)$ is itself a diffeomorphism between the open submanifold U and its coordinate image.

Example 1.38 (Graphs). If $f : U \subseteq \mathbb{R}^n \rightarrow \mathbb{R}^m$ is smooth, then its graph

$$\Gamma_f = \{(x, f(x)) : x \in U\}$$

is diffeomorphic to U by projection $(x, f(x)) \mapsto x$.

Example 1.39 ($\mathbb{R}/\mathbb{Z}$). Let $M = \mathbb{R}/\mathbb{Z}$ with $a \sim b$ if $b - a \in \mathbb{Z}$. Two chart domains may be chosen by deleting one equivalence class at a time, producing coordinates in open intervals of $\mathbb{R}$. These charts form a smooth atlas. The map

$$G : \mathbb{R}/\mathbb{Z} \rightarrow S^1, \quad [a] \mapsto (\cos 2\pi a, \sin 2\pi a)$$

is a diffeomorphism.

2 Tangent and Cotangent Vectors

2.1 Smooth Scalar Functions and Bump Functions

Let M be a smooth manifold and let $f : M \rightarrow \mathbb{R}$. Since $\mathbb{R}$ has its standard smooth structure, smoothness of f is especially simple.

Definition 2.1. A scalar function $f : M \rightarrow \mathbb{R}$ is smooth if for every $x \in M$ and chart (U, φ) with $x \in U$, the function

$$f \circ \varphi^{-1} : \varphi(U) \rightarrow \mathbb{R}$$

is smooth.

We write

$$C^\infty(M) = \{f : M \rightarrow \mathbb{R} : f \text{ is smooth}\}.$$

This is a real vector space.

2.1.1 Compactness

Definition 2.2 (Compact subset). A subset $K \subseteq X$ of a topological space is compact if every open cover of K admits a finite subcover.

Proposition 2.3. If X is Hausdorff, every compact subset $K \subseteq X$ is closed.

Proof. Fix $x \notin K$. For each $y \in K$, choose disjoint neighborhoods $U_y \ni x$ and $V_y \ni y$. By compactness, finitely many $V_{y_1}, \dots, V_{y_r}$ cover K . Then

$$U = U_{y_1} \cap \dots \cap U_{y_r}$$

is an open neighborhood of x disjoint from K . Hence $X \setminus K$ is open. $\square$

2.1.2 Bump functions

Define

$$f(t) = \begin{cases} e^{-1/t}, & t > 0, \\ 0, & t \leq 0. \end{cases}$$

Then $f \in C^\infty(\mathbb{R})$. Set

$$g(t) = \frac{f(t)}{f(t) + f(1-t)}.$$

Then g is smooth, equals 0 for $t \leq 0$, and equals 1 for $t \geq 1$. Define

$$h(t) = g(t+2)g(2-t).$$

Then h is smooth, equals 1 on $[-1, 1]$, and vanishes for $|t| \geq 2$.

In $\mathbb{R}^n$, define

$$H(x_1, \dots, x_n) = h(x_1) \cdots h(x_n).$$

This produces a smooth cutoff supported in a bounded region.

Using a chart (U, φ) about a point of a manifold and arranging a small Euclidean ball B with $\overline{B} \subseteq \varphi(U)$, one obtains a smooth function on M that is 1 near the chosen point and 0 outside U .

2.2 Partitions of Unity

Definition 2.4 (Support). For a function $f : M \rightarrow \mathbb{R}$, its support is

$$\text{supp}(f) = \overline{\{x \in M : f(x) \neq 0\}}.$$

Definition 2.5 (Partition of unity). Let $\{V_\alpha\}_{\alpha \in A}$ be an open cover of M . A collection of smooth functions $\{f_\alpha\}_{\alpha \in A}$ is a partition of unity subordinate to this cover if:

1. $f_\alpha \geq 0$;
2. for each $x \in M$, only finitely many $f_\alpha(x)$ are nonzero in a neighborhood of x (local finiteness);
3. $\text{supp}(f_\alpha) \subseteq V_\alpha$;
4. $\sum_{\alpha \in A} f_\alpha(x) = 1$ for every $x \in M$.

Theorem 2.6 (Existence of partitions of unity). Every open cover of a smooth Hausdorff second-countable manifold admits a smooth partition of unity subordinate to it.

The proof in the notes proceeds by first constructing a locally finite family of smooth nonnegative bump functions subordinate to the cover, then normalizing their locally finite sum.

Proposition 2.7 (Application). Let $A \subseteq M$ be closed and let $U \subseteq M$ be open with $A \subseteq U$. There exists $\psi \in C^\infty(M)$ such that

$$\psi|_A = 1, \quad \text{supp}(\psi) \subseteq U.$$

In particular, $\psi = 0$ on $M \setminus U$.

Proof. The two sets U and $M \setminus A$ form an open cover of M . Let f_0, f_1 be a subordinate partition of unity, with $\text{supp}(f_0) \subseteq U$ and $\text{supp}(f_1) \subseteq M \setminus A$. Then on A , $f_1 = 0$, hence $f_0 = 1$. Take $\psi = f_0$. $\square$

2.3 The Derivative of a Function and the Cotangent Space

Suppose first that $M = \mathbb{R}^n$ and $f : \mathbb{R}^n \rightarrow \mathbb{R}$. At $a \in \mathbb{R}^n$,

$$Df(a) = \sum_{i=1}^n \frac{\partial f}{\partial x_i}(a) dx_i.$$

The notes seek an intrinsic definition that does not depend on the chosen chart.

Let M be an n -manifold, $a \in M$, and (U, φ) a chart with

$$\varphi = (x_1, \dots, x_n).$$

For $f \in C^\infty(M)$, the coordinate function $f \circ \varphi^{-1}$ is smooth near $\varphi(a)$. If all first partial derivatives vanish at $\varphi(a)$ in one chart, then they vanish in every chart, by the chain rule and invertibility of the Jacobian of a coordinate change.

Definition 2.8. Let

$$Z_a = \{f \in C^\infty(M) : f \text{ has zero derivative at } a\}.$$

The cotangent space at a is

$$T_a^*M = C^\infty(M)/Z_a.$$

For $f \in C^\infty(M)$, denote its class by

$$(df)_a = [f]_{Z_a}.$$

This is called the derivative or differential of f at a .

Remark 2.9. Two smooth functions f and g have the same derivative at a exactly when $f - g$ has zero derivative at a .

Proposition 2.10. Let M be an n -manifold and $a \in M$.

1. T_a^*M is an n -dimensional real vector space.
2. If (U, φ) is a chart at a with $\varphi = (x_1, \dots, x_n)$, then

$$\{(dx_1)_a, \dots, (dx_n)_a\}$$

is a basis of T_a^*M .

3. For every $f \in C^\infty(M)$,

$$(df)_a = \sum_{i=1}^n \frac{\partial(f \circ \varphi^{-1})}{\partial x_i}(\varphi(a)) (dx_i)_a.$$

2.4 Tangent Space as the Dual and as Derivations

Definition 2.11 (Tangent space). The tangent space at $a \in M$ is the dual space

$$T_a M = (T_a^* M)^*.$$

Thus $T_a M$ is an n -dimensional real vector space.

If (U, φ) is a chart with coordinates $(x_1, \dots, x_n)$, define the dual basis

$$\left\{ \frac{\partial}{\partial x_1} \Big|_a, \dots, \frac{\partial}{\partial x_n} \Big|_a \right\}$$

by

$$\frac{\partial}{\partial x_i} \Big|_a (dx_j)_a = \delta_{ij}.$$

Definition 2.12 (Derivation). A derivation at a is an $\mathbb{R}$ -linear map

$$X_a : C^\infty(M) \rightarrow \mathbb{R}$$

satisfying the Leibniz rule

$$X_a(fg) = f(a)X_a(g) + g(a)X_a(f)$$

for all $f, g \in C^\infty(M)$.

The notes record three basic properties:

1. constants have zero derivative under a derivation;
2. if $f(a) = g(a) = 0$, then $X_a(fg) = 0$;
3. if $f = g$ on a neighborhood of a , then $X_a(f) = X_a(g)$.

Thus derivations are local.

Proposition 2.13. On $\mathbb{R}^n$, every derivation at a has the form

$$X_a = \sum_{i=1}^n c_i \frac{\partial}{\partial x_i} \Big|_a, \quad c_i = X_a(x_i).$$

Consequently, $\text{Der}_a(\mathbb{R}^n)$ is naturally isomorphic to $T_a \mathbb{R}^n$.

Proof. For $f \in C^\infty(\mathbb{R}^n)$, the fundamental theorem of calculus yields a decomposition

$$f(x) = f(a) + \sum_{i=1}^n (x_i - a_i) g_i(x),$$

with smooth g_i satisfying $g_i(a) = \frac{\partial f}{\partial x_i}(a)$. Applying X_a and using the Leibniz rule gives

$$X_a(f) = \sum_{i=1}^n X_a(x_i) \frac{\partial f}{\partial x_i}(a).$$

□

For an arbitrary manifold, the same argument in a chart gives

$$T_a M \cong \text{Der}_a(M).$$

2.4.1 Change of coordinates

Suppose (U, φ) and (V, ψ) are charts with coordinates

$$\varphi = (x_1, \dots, x_n), \quad \psi = (y_1, \dots, y_n).$$

Then

$$\left. \frac{\partial}{\partial y_j} \right|_a = \sum_{i=1}^n \frac{\partial x_i}{\partial y_j}(\psi(a)) \left. \frac{\partial}{\partial x_i} \right|_a,$$

while

$$(dy_j)_a = \sum_{i=1}^n \frac{\partial y_j}{\partial x_i}(\varphi(a)) (dx_i)_a.$$

Thus the tangent and cotangent bases transform by inverse Jacobian matrices.

2.5 Derivatives of Smooth Maps

Let $F : M \rightarrow N$ be smooth and let $a \in M$.

Definition 2.14 (Derivative / pushforward). The derivative of F at a is the linear map

$$DF_a = F_{*,a} : T_a M \rightarrow T_{F(a)} N$$

defined by

$$DF_a(X_a)(f) = X_a(f \circ F), \quad f \in C^\infty(N).$$

If (U, φ) is a chart on M with coordinates $x_1, \dots, x_m$ and (V, ψ) is a chart on N with coordinates $y_1, \dots, y_n$, then the matrix of DF_a with respect to the coordinate bases is

$$\text{Jac}(\psi \circ F \circ \varphi^{-1})(\varphi(a)).$$

Equivalently,

$$DF_a \left(\frac{\partial}{\partial x_i} \Big|_a \right) = \sum_{j=1}^n \frac{\partial(y_j \circ F)}{\partial x_i}(a) \frac{\partial}{\partial y_j} \Big|_{F(a)}.$$

Proposition 2.15 (Basic properties). For smooth maps $F : M \rightarrow N$ and $G : N \rightarrow P$:

1. $DF_a : T_a M \rightarrow T_{F(a)} N$ is linear;
2. $D(G \circ F)_a = DG_{F(a)} \circ DF_a$;
3. $D(\text{id}_M)_a = \text{id}_{T_a M}$;
4. if F is a diffeomorphism, then DF_a is an isomorphism and

$$(DF_a)^{-1} = D(F^{-1})_{F(a)};$$

5. if F is constant on a connected manifold, then $DF_a = 0$ for every a , and conversely a smooth map with zero derivative everywhere is constant on each connected component.

2.5.1 Tangent vectors as velocities of curves

Definition 2.16 (Smooth curve). A smooth curve on M is the image of a smooth map

$$\gamma : (-\varepsilon, \varepsilon) \rightarrow M.$$

Let $a = \gamma(0)$. Its velocity at time t_0 is

$$\gamma'(t_0) = \gamma_{*,t_0} \left(\frac{d}{dt} \Big|_{t_0} \right) \in T_{\gamma(t_0)} M.$$

In local coordinates $x_1, \dots, x_n$,

$$\gamma'(t_0) = \sum_{i=1}^n \frac{d(x_i \circ \gamma)}{dt}(t_0) \frac{\partial}{\partial x_i} \Big|_{\gamma(t_0)}.$$

Every tangent vector is the velocity of some smooth curve through the point.

Example 2.17 (Matrix groups). For $A \in M_{n \times n}(\mathbb{R})$ and $H \in M_{n \times n}(\mathbb{R})$, the curve

$$\gamma(t) = A + tH$$

has velocity H at $t = 0$. Thus

$$T_A M_{n \times n}(\mathbb{R}) \cong M_{n \times n}(\mathbb{R}).$$

For $A \in \text{GL}(n, \mathbb{R})$, since $\text{GL}(n, \mathbb{R})$ is open in $M_{n \times n}(\mathbb{R})$,

$$T_A \text{GL}(n, \mathbb{R}) \cong M_{n \times n}(\mathbb{R}).$$

The Lie algebra of a Lie group G is $T_e G$.

2.5.2 Tangent spaces of regular level sets

Theorem 2.18. Let $F : M^n \rightarrow N^m$ be smooth, let $c \in N$, and suppose DF_a is surjective for every $a \in F^{-1}(c)$. Then $F^{-1}(c)$ is a smooth manifold of dimension $n - m$, and for every $a \in F^{-1}(c)$,

$$T_a F^{-1}(c) = \ker DF_a.$$

Example 2.19 (Orthogonal group). Let

$$\text{O}(n) = \{A \in M_{n \times n}(\mathbb{R}) : AA^T = I\}.$$

Consider $F(A) = AA^T$. At the identity,

$$DF_I(H) = H + H^T.$$

Hence

$$T_I \text{O}(n) = \{H : H + H^T = 0\},$$

the skew-symmetric matrices.

Example 2.20 (Tangent space to a sphere). For $S^n = \{x \in \mathbb{R}^{n+1} : \|x\|^2 = 1\}$ and $a \in S^n$,

$$DF_a(v) = 2a \cdot v.$$

Therefore

$$T_a S^n = \{v \in \mathbb{R}^{n+1} : a \cdot v = 0\}.$$

Geometrically, the tangent space is the hyperplane orthogonal to the radius vector a .

3 Immersions, Submersions, and Submanifolds

3.1 Immersions and Submersions

Definition 3.1. Let $F : M^m \rightarrow N^n$ be smooth and $p \in M$.

- F is an *immersion at p* if $DF_p : T_p M \rightarrow T_{F(p)} N$ is injective.
- F is a *submersion at p* if DF_p is surjective.

Canonical examples are the inclusion

$$\iota : \mathbb{R}^m \hookrightarrow \mathbb{R}^n, \quad (x_1, \dots, x_m) \mapsto (x_1, \dots, x_m, 0, \dots, 0),$$

for $m \leq n$, and the projection

$$\pi : \mathbb{R}^n \rightarrow \mathbb{R}^m, \quad (x_1, \dots, x_n) \mapsto (x_1, \dots, x_m),$$

for $m \leq n$.

Theorem 3.2 (Local normal form). If F is an immersion or a submersion at p , then one can choose charts near p and $F(p)$ so that the coordinate representation is respectively the canonical inclusion or canonical projection.

Definition 3.3 (Embedding). A smooth map $F : M \rightarrow N$ is an embedding if it is an immersion and a homeomorphism from M onto its image $F(M)$, where the image has the subspace topology.

Example 3.4. Every diffeomorphism is an embedding. The canonical inclusion $\mathbb{R}^m \hookrightarrow \mathbb{R}^n$ is an embedding.

Example 3.5 (Immersion that is not an embedding). The notes give a self-intersecting plane curve as an immersion whose image has a crossing. Although the derivative has rank 1 everywhere, the map is not one-to-one and therefore is not an embedding.

Remark 3.6. Even a one-to-one immersion need not be an embedding if the topology induced on the image does not agree with the original topology of the domain.

Example 3.7 (Graph embedding). If $f : W \subseteq M \rightarrow N$ is smooth, then

$$F : W \rightarrow M \times N, \quad p \mapsto (p, f(p))$$

is an embedding. Therefore the graph of a smooth function is a submanifold of $M \times N$.

3.2 Submanifolds

Definition 3.8 (Embedded submanifold). Let M be a smooth m -manifold. A subset $S \subseteq M$ is a k -dimensional submanifold if for every $p \in S$ there exists a chart (U, φ) of M with $p \in U$ such that

$$\varphi(U \cap S) = \{x \in \varphi(U) : x_{k+1} = \cdots = x_m = 0\}.$$

The subspace topology on S , together with the coordinate maps obtained by restricting φ and projecting onto the first k coordinates, gives S its induced smooth structure.

Proposition 3.9. If S is a submanifold of M , then the inclusion map

$$i : S \hookrightarrow M$$

is an embedding.

Remark 3.10. Conversely, if $i : S \rightarrow M$ is an embedding, then $i(S)$ is a submanifold of M . Thus embedded submanifolds are precisely images of embeddings.

Example 3.11. The coordinate plane

$$\mathbb{R}^k \times \{0\} \subseteq \mathbb{R}^m$$

is a k -dimensional submanifold.

Example 3.12 (Graphs). If $f : U \subseteq \mathbb{R}^m \rightarrow \mathbb{R}^n$ is smooth, then

$$\Gamma_f = \{(x, f(x)) : x \in U\}$$

is an m -dimensional submanifold of $\mathbb{R}^{m+n}$.

Remark 3.13. The graph of a nonsmooth function may be a smooth manifold

as an abstract manifold but need not be a submanifold in the given ambient coordinates. The notes use $y = |x|$ as the basic example: it admits a global parameterization by x , but at the corner it is not locally a smooth embedded curve in $\mathbb{R}^2$.

Definition 3.14 (Immersed submanifold). A subset $S \subseteq M$ is a k -dimensional immersed submanifold if $S = f(\tilde{S})$ for some k -manifold $\tilde{S}$ and immersion $f : \tilde{S} \rightarrow M$.

Example 3.15. A self-intersecting immersed curve in $\mathbb{R}^2$ is an immersed one-dimensional submanifold, though its image need not be an embedded submanifold.

4 Smooth Vector Fields and Differential Forms

4.1 The Tangent Bundle

Let M be an m -dimensional manifold. For each $p \in M$, let $T_p M$ be its tangent space. Define the tangent bundle

$$TM = \bigsqcup_{p \in M} T_p M.$$

There is a natural projection

$$\pi : TM \rightarrow M, \quad X_p \mapsto p.$$

If $(U_\alpha, \varphi_\alpha)$ is a chart on M with coordinates $x^1, \dots, x^m$, then every $X_p \in T_p M$ may be written

$$X_p = \sum_{i=1}^m a^i \left. \frac{\partial}{\partial x^i} \right|_p.$$

This gives a chart on TM by

$$\tilde{\varphi}_\alpha(X_p) = (\varphi_\alpha(p), a^1, \dots, a^m) \in \mathbb{R}^{2m}.$$

Thus TM is a smooth manifold of dimension $2m$.

Definition 4.1 (Vector field). A tangent vector field on M is a map

$$X : M \rightarrow TM$$

such that $X(p) \in T_p M$ for every p . Equivalently,

$$\pi \circ X = \text{id}_M.$$

Definition 4.2 (Smooth vector field). A vector field $X : M \rightarrow TM$ is smooth if it is smooth as a map of manifolds.

We write $\mathfrak{X}(M)$ for the set of all smooth vector fields on M .
In local coordinates,

$$X|_U = \sum_{i=1}^m a^i \frac{\partial}{\partial x^i}.$$

The notes record the basic criterion:

Proposition 4.3. A vector field X is smooth on U if and only if its coefficient functions $a^1, \dots, a^m$ are smooth.

Example 4.4. On $\mathbb{R}^2$ with coordinates (x, y) ,

$$X = e^x \sin y \frac{\partial}{\partial x} + ye^{x^2} \frac{\partial}{\partial y}$$

is smooth.

4.2 Smooth Vector Fields as Derivations

A smooth vector field acts on smooth functions pointwise:

$$X(f)(p) = X_p(f), \quad f \in C^\infty(M).$$

Thus

$$X : C^\infty(M) \rightarrow C^\infty(M)$$

is $\mathbb{R}$ -linear and satisfies

$$X(fg) = X(f)g + fX(g).$$

Hence smooth vector fields are global derivations of the algebra $C^\infty(M)$.

4.3 Differential One-Forms

Definition 4.5 (Smooth one-form). A smooth one-form on M is a smooth section of the cotangent bundle:

$$\omega : M \rightarrow T^*M, \quad \omega(p) \in T_p^*M.$$

In a coordinate chart $(U, x^1, \dots, x^m)$,

$$\omega|_U = \sum_{i=1}^m a_i dx^i,$$

where the coefficient functions a_i are smooth. Equivalently, ω acts on smooth vector fields by a smooth scalar function:

$$\omega : \mathfrak{X}(M) \rightarrow C^\infty(M), \quad X \mapsto \omega(X),$$

where $\omega(X)(p) = \omega_p(X_p)$.

Example 4.6. On $\mathbb{R}^3$, the forms dx, dy, dz are smooth. For example,

$$\omega = z^2 dx - 4y dz$$

is a smooth one-form.

4.3.1 Change of coordinates

If $(x^1, \dots, x^m)$ and $(y^1, \dots, y^m)$ are two coordinate systems, then

$$\frac{\partial}{\partial x^i} = \sum_j \frac{\partial y^j}{\partial x^i} \frac{\partial}{\partial y^j}, \quad dy^j = \sum_i \frac{\partial y^j}{\partial x^i} dx^i.$$

Hence if

$$\omega = \sum_i a_i dx^i = \sum_j b_j dy^j,$$

then on overlaps

$$a_i = \sum_j b_j \frac{\partial y^j}{\partial x^i}.$$

Example 4.7 (Differential of a function). For $f \in C^\infty(M)$,

$$(df)_p = \sum_i \frac{\partial f}{\partial x^i}(p) (dx^i)_p.$$

These local expressions agree under coordinate changes, so df is a globally defined smooth one-form.

4.4 Pullbacks

Let $F : M \rightarrow N$ be smooth.

Definition 4.8 (Pullback of a one-form). The pullback

$$F^* : \Omega^1(N) \rightarrow \Omega^1(M)$$

is defined by

$$(F^*\omega)_p(X_p) = \omega_{F(p)}(F_{*,p}X_p).$$

In local coordinates x^i on M , y^j on N , if

$$\omega = \sum_j a_j dy^j,$$

then

$$F^*\omega = \sum_j (a_j \circ F) d(y^j \circ F).$$

Example 4.9. Let

$$F : \mathbb{R}^3 \rightarrow \mathbb{R}^2, \quad (x_1, x_2, x_3) \mapsto (2x_1 + x_2, x_3^2),$$

and let

$$\omega = e^{2y_1+y_2} dy_1 + dy_2.$$

Then

$$F^*\omega = e^{2(2x_1+x_2)+x_3^2} d(2x_1 + x_2) + d(x_3^2).$$

4.5 k -Forms

For each $p \in M$, the space of alternating k -linear forms on $T_p M$ is

$$\Lambda^k T_p^* M.$$

A smooth k -form is a smooth section of the bundle $\Lambda^k T^* M$.

In coordinates,

$$\omega|_U = \sum_{i_1 < \dots < i_k} a_{i_1 \dots i_k} dx^{i_1} \wedge \dots \wedge dx^{i_k},$$

with smooth coefficient functions $a_{i_1 \dots i_k}$.

Equivalently, a k -form is an alternating $C^\infty(M)$ -multilinear map

$$\omega : \mathfrak{X}(M)^k \rightarrow C^\infty(M).$$

Example 4.10. On $\mathbb{R}^3$,

$$dx \wedge dy$$

is a smooth two-form, and

$$(x^2y - 3z) dx \wedge dy \wedge dz$$

is a smooth three-form.

If $\omega = a dx^1 \wedge \cdots \wedge dx^n$ and also

$$\omega = b dy^1 \wedge \cdots \wedge dy^n,$$

then on an overlap

$$a = b \det \left(\frac{\partial y}{\partial x} \right).$$

4.6 Operations on Differential Forms

4.6.1 Pullback of k -forms

For $F : M \rightarrow N$ and $\omega \in \Omega^k(N)$,

$$(F^*\omega)_p(v_1, \dots, v_k) = \omega_{F(p)}(F_{*,p}v_1, \dots, F_{*,p}v_k).$$

In local coordinates,

$$F^* \left(\sum_I a_I dy^I \right) = \sum_I (a_I \circ F) d(y^{i_1} \circ F) \wedge \cdots \wedge d(y^{i_k} \circ F).$$

4.6.2 Exterior derivative

Definition 4.11. The exterior derivative is the operator

$$d : \Omega^k(M) \rightarrow \Omega^{k+1}(M)$$

defined in local coordinates by

$$\begin{aligned} \omega &= \sum_{i_1 < \cdots < i_k} a_{i_1 \dots i_k} dx^{i_1} \wedge \cdots \wedge dx^{i_k} \\ \implies d\omega &= \sum_{i_1 < \cdots < i_k} da_{i_1 \dots i_k} \wedge dx^{i_1} \wedge \cdots \wedge dx^{i_k}. \end{aligned}$$

For $f \in C^\infty(M)$,

$$df = \sum_i \frac{\partial f}{\partial x^i} dx^i.$$

4.7 Orientation and Integration

On $\mathbb{R}^n$, if f has compact support, one writes

$$\int_{\mathbb{R}^n} f(x) dx^1 \cdots dx^n.$$

For an n -form

$$\omega = f dx^1 \wedge \cdots \wedge dx^n,$$

define

$$\int_{\mathbb{R}^n} \omega = \int_{\mathbb{R}^n} f(x) dx^1 \cdots dx^n.$$

Under a coordinate change $x = x(y)$,

$$dx^1 \wedge \cdots \wedge dx^n = \det \left(\frac{\partial x}{\partial y} \right) dy^1 \wedge \cdots \wedge dy^n.$$

To obtain a consistent global sign, the manifold must be orientable.

Definition 4.12 (Orientable manifold). A smooth manifold M is orientable if it admits an atlas $\{(U_\alpha, \varphi_\alpha)\}$ such that on every nonempty overlap,

$$\det D(\varphi_\alpha \circ \varphi_\beta^{-1}) > 0.$$

Such an atlas is called oriented.

For an orientable manifold and a compactly supported top-degree form ω , choose an oriented atlas and a partition of unity $\{f_\alpha\}$ subordinate to the chart domains. Then define

$$\int_M \omega = \sum_\alpha \int_{\varphi_\alpha(U_\alpha)} (\varphi_\alpha^{-1})^* (f_\alpha \omega).$$

4.8 Smooth Maps and Vector Fields

Let $F : M \rightarrow N$ be smooth and let X be a vector field on M . At each point,

$$F_{*,p}(X_p) \in T_{F(p)}N.$$

In general this need not define a vector field on all of N : if F is not injective, two points in the same fiber may push forward to different vectors; if F is not surjective, there may be no vector assigned outside $F(M)$.

Definition 4.13 (F -related vector fields). Let $X \in \mathfrak{X}(M)$ and $Y \in \mathfrak{X}(N)$. They are F -related if

$$F_{*,p}(X_p) = Y_{F(p)}$$

for every $p \in M$.

Proposition 4.14. The vector fields X and Y are F -related if and only if

$$X(f \circ F) = (Yf) \circ F$$

for every $f \in C^\infty(N)$.

Example 4.15. Let

$$F : \mathbb{R} \rightarrow \mathbb{R}^2, \quad t \mapsto (\cos t, \sin t).$$

Then

$$X = \frac{d}{dt}$$

is F -related to

$$Y = -y \frac{\partial}{\partial x} + x \frac{\partial}{\partial y}.$$

Proposition 4.16 (Pushforward by a diffeomorphism). If $F : M \rightarrow N$ is a diffeomorphism and $X \in \mathfrak{X}(M)$, there is a unique vector field $F_*X \in \mathfrak{X}(N)$ that is F -related to X . It is given by

$$(F_*X)_y = F_{*,F^{-1}(y)}(X_{F^{-1}(y)}).$$

If $F : M \rightarrow N$ and $G : N \rightarrow P$ are diffeomorphisms, then

$$(G \circ F)_*X = G_*(F_*X).$$

Also, pushforward preserves Lie brackets:

$$F_*[X, X'] = [F_*X, F_*X'].$$

In coordinates, if

$$X = \sum_i a_i \frac{\partial}{\partial x_i},$$

then on a chart of N with coordinates y_j ,

$$F_*X = \sum_j \left(\sum_i (a_i \circ F^{-1}) \frac{\partial(y_j \circ F)}{\partial x_i} \circ F^{-1} \right) \frac{\partial}{\partial y_j}.$$

4.9 Integral Curves and Flows

Let $X \in \mathfrak{X}(U)$ be a smooth vector field on an open subset $U \subseteq M$.

Definition 4.17 (Integral curve). An integral curve of X through $p \in U$ is a smooth curve

$$\gamma : (-\varepsilon, \varepsilon) \rightarrow M$$

such that

$$\gamma(0) = p, \quad \gamma'(t) = X_{\gamma(t)}$$

for all t in its interval of definition.

The equation

$$\gamma'(t) = X(\gamma(t))$$

is an ordinary differential equation with smooth coefficients. Locally, standard ODE theory gives existence and uniqueness.

Example 4.18 (Constant coordinate field). On $\mathbb{R}^2$, let

$$X = \frac{\partial}{\partial x_1}.$$

Writing $\gamma(t) = (x_1(t), x_2(t))$, the integral-curve equation becomes

$$x_1'(t) = 1, \quad x_2'(t) = 0.$$

Hence

$$\gamma(t) = (t + c_1, c_2) = p + t(1, 0).$$

Similarly, the integral curves of $\frac{\partial}{\partial x_2}$ are vertical lines.

Example 4.19 (Radial field). Let

$$X = x_1 \frac{\partial}{\partial x_1} + x_2 \frac{\partial}{\partial x_2}.$$

Then

$$x_1' = x_1, \quad x_2' = x_2,$$

so

$$\gamma(t) = e^t p.$$

Integral curves move along rays from the origin.

Example 4.20 (Rotation field). Let

$$X = -x_2 \frac{\partial}{\partial x_1} + x_1 \frac{\partial}{\partial x_2}.$$

Then

$$x_1' = -x_2, \quad x_2' = x_1.$$

For $p = (c_1, c_2)$, the solution is circular motion; equivalently, in polar form,

$$\gamma(t) = r(\cos(t + \theta_0), \sin(t + \theta_0)).$$

Thus the integral curves are circles centered at the origin.